

cl_01_distance

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Purpose

The goal of this code here is to do two things (1) this identifies the centroid of each county (2) this identifies the distance from the centroid of each county to each of the foodhub identified in the foodhub directory

Get the centroid of each of the county

```
# step 1: we identify all the counities in US

us_counties <- counties(cb = TRUE, year = 2023)

# Step 2: identifies the centroid
# we use 5070 since it stands for NAD83, an Albers Equal Area projection designed for the continental U
# We use 4326 since it stands for WGS84, the plain longitude/latitude system in degrees so that we expr
county_centroid <- us_counties %>%
  st_transform(5070) %>%
  st_centroid() %>%
  mutate(x = st_coordinates(geometry)[, 1],
         y = st_coordinates(geometry)[, 2])
```

```
## Warning: st_centroid assumes attributes are constant over geometries
```

```
# this will express the centroid by meter and we use 4326 to transform it into the x-y coordinates

county_centroid_xy <- county_centroid %>%
  st_transform(4326) %>%
  mutate(x = st_coordinates(geometry)[, 1],
         y = st_coordinates(geometry)[, 2])
```

Validate by mapping one of the county and the centroid for it

```
# step 1 : set the seed
set.seed(0307)

# step 2: randomly select a county from our us_county dataset
random_county <- sample(us_counties$GEOID, 1)
```

```
# subset both the county and the centroid for the sample

poly_sample <- us_counties %>% filter(GEOID == random_county) %>% st_transform(5070)
cent_sample <- county_centroid %>% filter(GEOID == random_county)

# Step 3: map the county and its centroid

#mapview(poly_sample) + mapview(cent_sample, color = "red")
```

Step 2

Load the dataset from the foodhub directory and calculate the distance from the county centroid to each of the foodhub

note: if foodhub does not exist in this county, we will map the distance to the closest foodhub around

```
foodhub_raw <- read_excel("../data/source/foodhub_2026-75161420.xlsx")

foodhub_directory <- foodhub_raw %>%
  select(listing_id, listing_name, location_x, location_y, products) %>%
  rename(lon = location_x, lat = location_y) %>%
  mutate(lon = as.numeric(lon),
         lat = as.numeric(lat)) %>%
  filter(!is.na(lon), !is.na(lat))

#glimpse(foodhub_directory)
```

validation: we convert the location into spatial and view it in map

```
hubs_sf <- st_as_sf(foodhub_directory, coords = c("lon", "lat"), crs = 4326)

#mapview(hubs_sf)

# Also lookup which county this foodhub actually locates

county_lookup <- us_counties %>%
  select(STATEFP, GEOID, county_name = NAME) %>% # select() can rename in-place
  st_transform(4326)

hubs_geo <- st_join(hubs_sf, county_lookup, join = st_within)
```

convert the xy coordinate of the hubs into meters by 5070 so that we can get the distance from the centroid to the hub

```
hub_proj <- st_transform(hubs_sf, 5070)

#check whether two are in the same scale
st_crs(hub_proj) == st_crs(county_centroid)
```

```
## [1] TRUE
```

******identify the distance between the centroid and the hub

```
distance_hub <- st_distance(county_centroid, hub_proj)

# (number of counties) x (number of hubs)
#dim(distance_hub)
```

for each of the row in the distance matrix, identify the hub that is closest to each of the county and the distance

```
nearest_dist <- apply(distance_hub, 1, min)
nearest_idx <- apply(distance_hub, 1, which.min)
```

in the county dataset, for each of the county centroid, identify the name and the distance to the closest hub

```
hub_distance <- county_centroid %>%
  select(GEOID, STATE_NAME, NAME, STATEFP) %>%
  mutate(
    `distance_nearest_hub(mile)` = set_units(set_units(nearest_dist, "m"), "mi") %>% as.numeric(),
    nearest_hub_name = foodhub_directory$listing_name[nearest_idx],
    nearest_hub_id = foodhub_directory$listing_id[nearest_idx]) %>%
  rename(county_name = "NAME",
         state = "STATE_NAME")
```

Merge this with other hub information

```
hub_attrs <- hubs_geo %>%
  st_drop_geometry() %>%
  select(listing_id,
         hub_county_geoid = GEOID,
         hub_county_name = county_name,
         hub_products = products)

hub_distance_full <- hub_distance %>%
  left_join(hub_attrs, by = c("nearest_hub_id" = "listing_id")) %>%
  mutate(hub_desert = (`distance_nearest_hub(mile)` >= 20))

# identify the places that are out of the continental US
hub_distance_full <- hub_distance_full %>%
  filter(STATEFP <= 56) %>%
  select(-STATEFP)
```

save the dataset

```
write.csv(hub_distance_full, "../data/outcome/hub_distance_full.csv")
```